SYSTEMS AND METHODS FOR FULL-BODY CONTEXTUAL MONITORING OF INTIMATE PHYSIOLOGY USING A WEARABLE RING OR CHEST-PATCH ARCHITECTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to previously filed provisional applications directed to multi-node physiological monitoring architectures, anatomical geometry acquisition systems, and computational compatibility modeling platforms, including but not limited to systems describing wearable penile physiology monitoring devices, anatomical geometry scanners, and intraluminal compliance measurement systems. The present disclosure may operate independently or as part of an integrated multi-node biometric ecosystem configured to generate synchronized physiological context data across multiple body regions.

FIELD OF THE INVENTION

The present disclosure relates generally to wearable biometric sensing systems and physiological context monitoring technologies. More specifically, the disclosure pertains to a full-body contextual physiology monitoring device configured as a wearable ring, band, patch, or chest-mounted module designed to capture systemic physiological signals relevant to intimate health, arousal context, autonomic state regulation, and whole-body biometric correlation.

SUMMARY OF THE DISCLOSURE

In various embodiments, the disclosed device (hereafter “Node 5”) functions as a systemic context acquisition platform configured to measure physiological variables outside localized anatomical sensing regions. The system provides environmental and autonomic reference data used to interpret localized intimate physiology measurements collected by companion sensing nodes.

The wearable architecture may be implemented as:

- a finger-worn smart ring;
- a wrist-mounted wearable;
- a chest patch positioned over the sternum or cardiac region;
- an adhesive biosensing patch;
- or other body-coupled wearable configurations.

The system may incorporate one or more sensing modalities including photoplethysmography (PPG), electrocardiography (ECG), temperature sensing, electrodermal activity sensing, respiratory motion sensing, accelerometry, gyroscopic orientation detection, magnetometer referencing, and multi-parameter autonomic signal fusion.

Collected data may be processed locally or transmitted to a computational platform configured to generate contextual physiology models describing autonomic balance, sympathetic and parasympathetic state transitions, stress response patterns, circulatory readiness, recovery status, and physiological synchronization relative to intimate events.